Telmisartan Improves MASLD by Activating Nrf2/HO-1 and SIRT1/AMPK, Significantly Reducing Oxidative Stress, Liver Enzymes, and Fibrosis.

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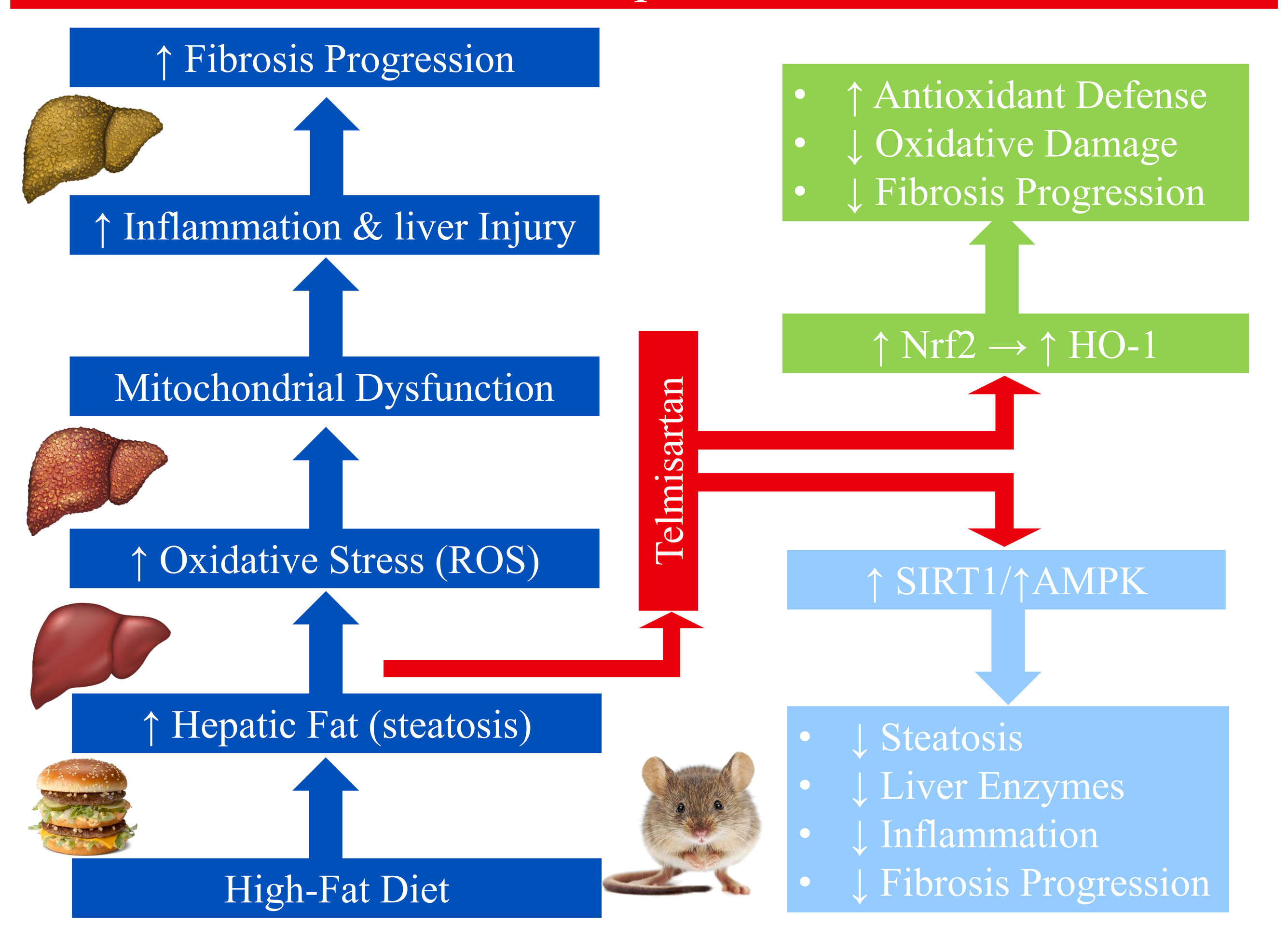
Background

Metabolic dysfunction–associated steatotic liver disease (MASLD) is now one of the most common chronic liver diseases, and it often travels with metabolic problems like obesity and type 2 diabetes. A big part of what drives MASLD is fat building up in the liver, which then triggers inflammation and oxidative stress. When oxidative stress builds up, mitochondria start to function poorly, creating even more reactive oxygen species and pushing the liver from simple fat accumulation toward more severe injury.
Telmisartan is best known as a blood pressure medication (an angiotensin receptor blocker), but it also has metabolic and anti-inflammatory effects that may be useful in MASLD. Prior work suggests it can support antioxidant defenses especially through pathways like Nrf2/HO-1 making it a promising candidate to limit oxidative damage and improve liver health in MASLD.

Telmisartan Improves MASLD



Objective and Methods

Objective
To determine whether telmisartan improves high-fat diet–induced MASLD and to evaluate whether its effects are associated with mitochondrial regulation (SIRT1/AMPK) and antioxidant signaling (Nrf2/HO-1).

Study design
- Animal model: Male Wistar rats
- Duration: 12 weeks total

Groups (n=6 per group; 4 groups, total n=24)
1. Control: standard chow
2. Telmisartan only: standard chow + telmisartan
3. HFD (MASLD model): high-fat diet for 12 weeks
4. HFD + Telmisartan: high-fat diet for 12 weeks + telmisartan during the final 4 weeks

Intervention
- Telmisartan dose: 10 mg/kg/day (given during weeks 9–12)

Outcomes measured
- Metabolic/lipid profile: body weight, glucose/insulin indices, total cholesterol, triglycerides
- Liver injury: ALT, AST
- Oxidative stress & antioxidants: MDA, GSH, SOD, catalase
- Inflammation: hepatic TNF-α, IL-1β
- Histology: liver steatosis and fibrosis assessment
- Mechanistic signaling: protein expression of SIRT1/AMPK, Nrf2/HO-1, plus markers of mitochondrial function/oxidative phosphorylation

Results

Telmisartan improves lipid and insulin markers in MASLD

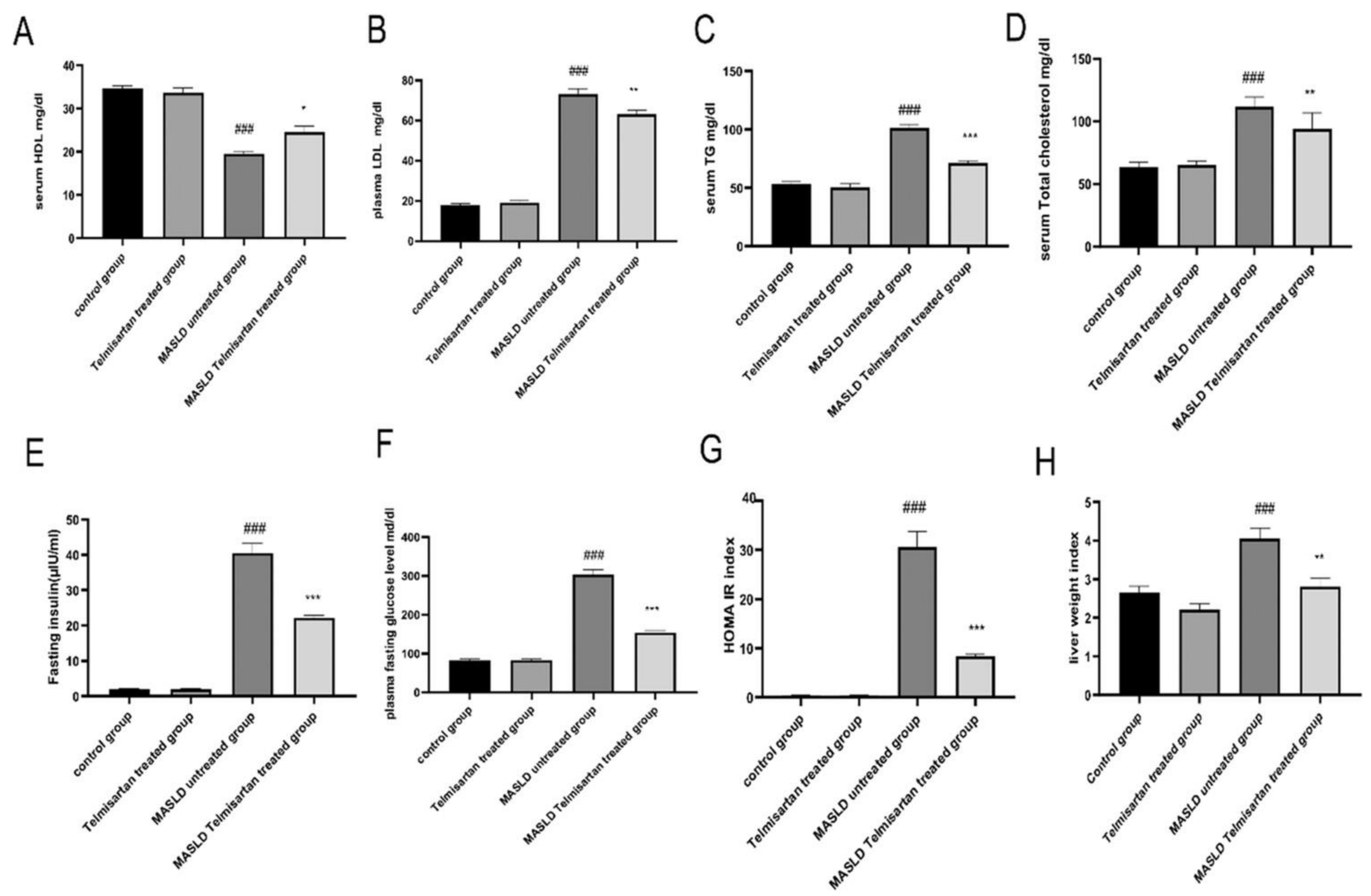


Figure 1. Effects of experimental groups on (A) HDL, (B) LDL, (C) TG, (D) total cholesterol, (E) fasting insulin, (F) fasting glucose, (G) HOMA-IR, and (H) liver weight ratio in HFD-induced MASLD rats. Data are mean ± SEM. One-way ANOVA with Tukey post hoc. *p < 0.05 vs Control; #p < 0.05 vs MASLD.

Telmisartan reduces AST and ALT in MASLD rats

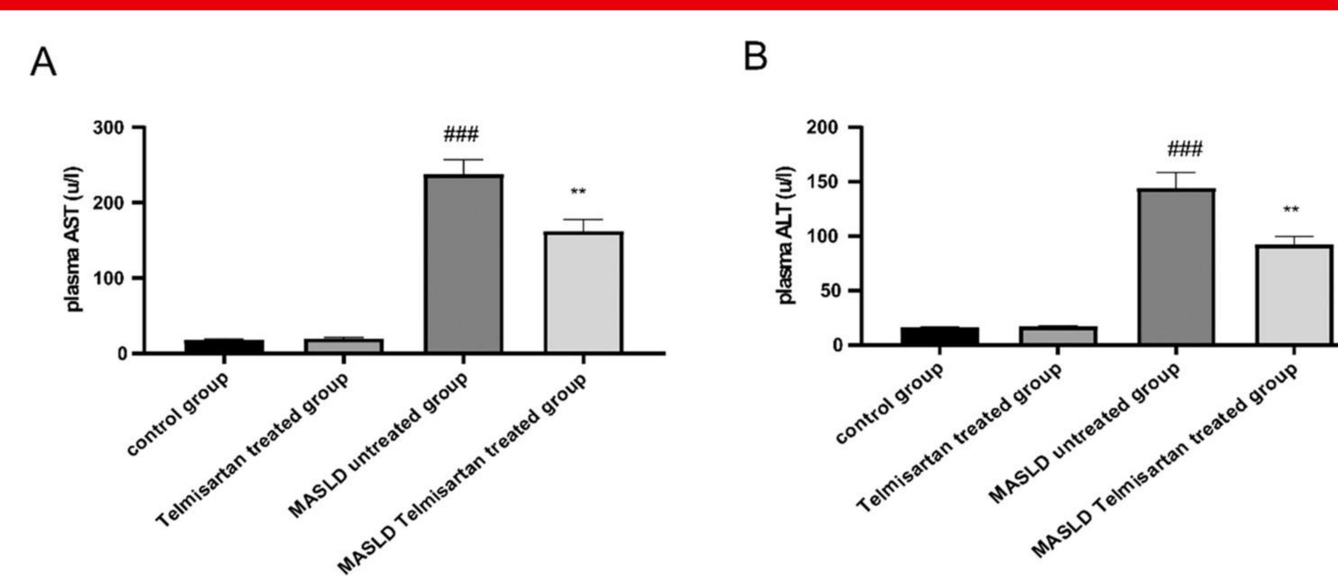


Figure 2. Telmisartan (TEL) effects on liver enzymes in HFD-induced MASLD rats: (A) plasma AST and (B) plasma ALT. Data are mean ± SEM. One-way ANOVA with Tukey post hoc. #p < 0.05 vs control; *p < 0.05 vs MASLD.

Telmisartan modulates hepatic oxidative stress markers in MASLD

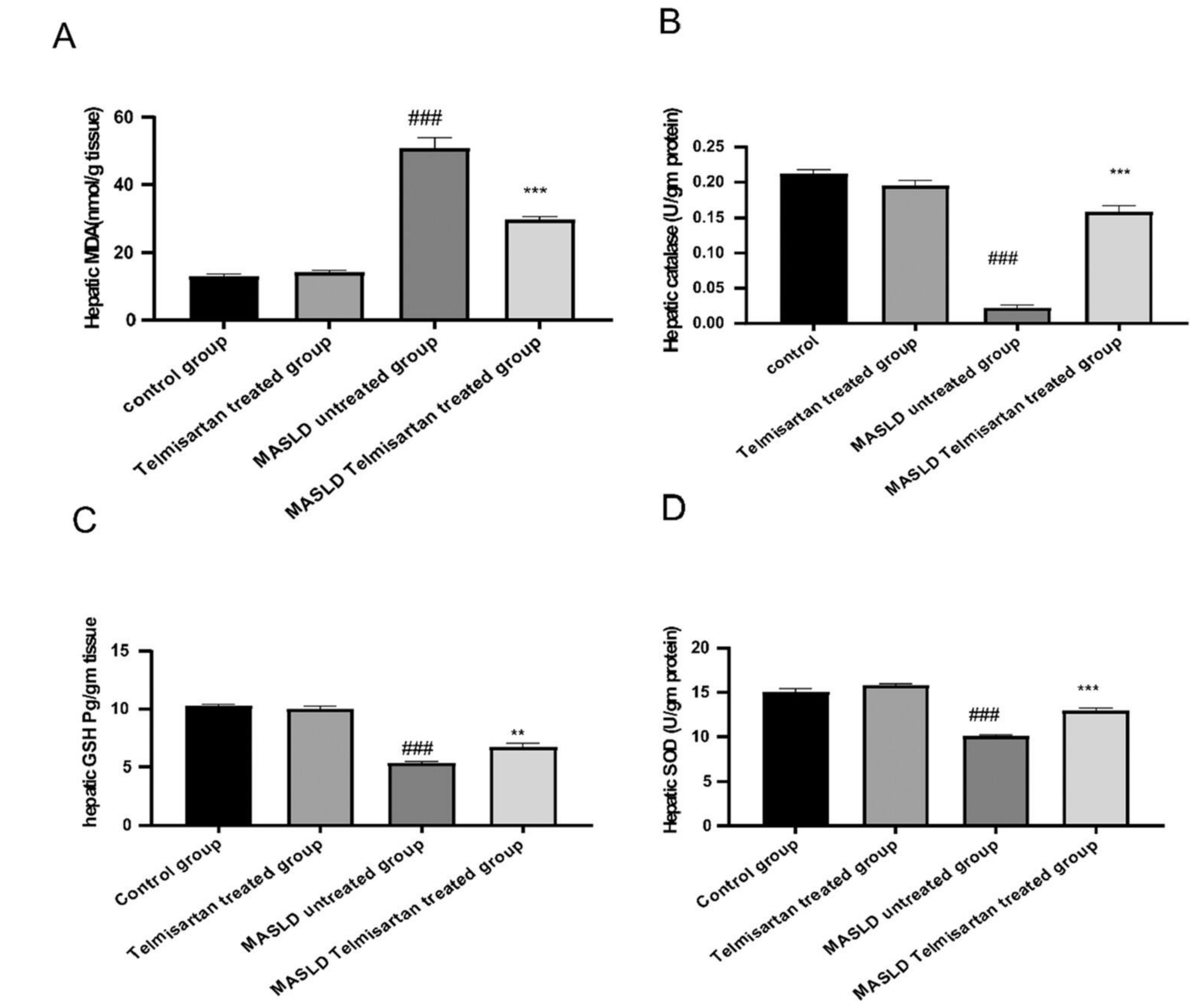


Figure 3. Effects of telmisartan (TEL) on hepatic oxidant/antioxidant status in HFD-induced MASLD: (A) MDA, (B) catalase, (C) GSH, and (D) SOD. Data are mean ± SEM. One-way ANOVA with Tukey post hoc. #p < 0.05 vs control; *p < 0.05 vs MASLD.

Telmisartan reduces hepatic TNF-α and IL-6 in MASLD

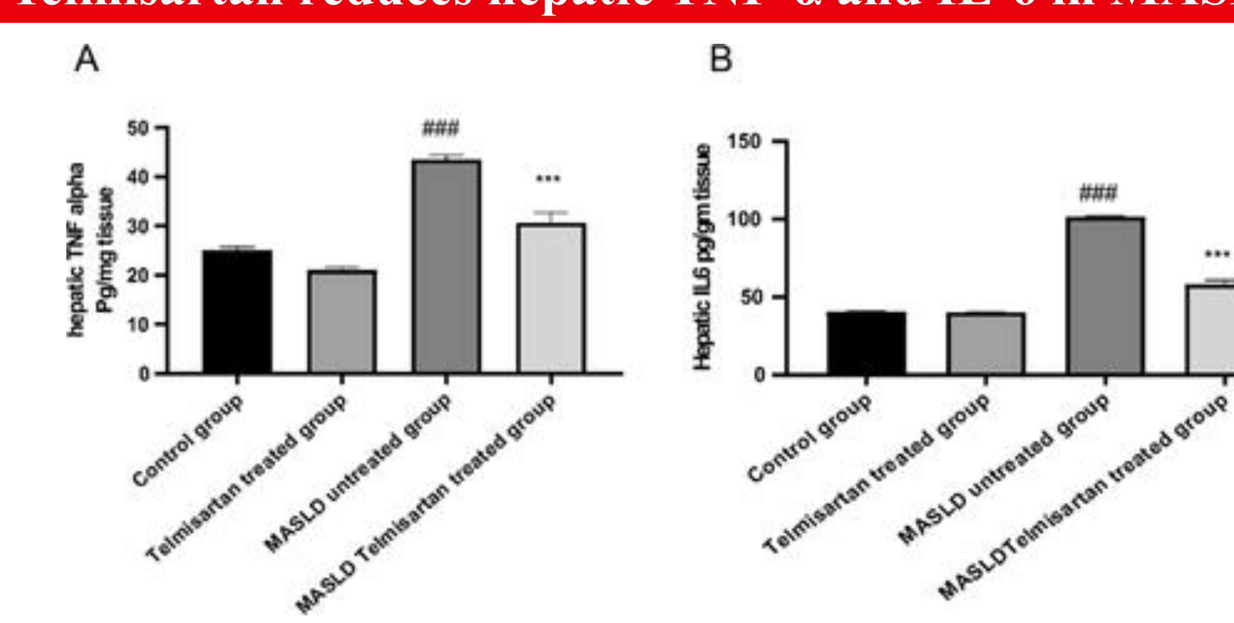


Figure 4. Effects of telmisartan (TEL) on hepatic inflammatory markers in HFD-induced MASLD: (A) TNF-α and (B) IL-6. Data are mean ± SEM. One-way ANOVA with Tukey post hoc. #p < 0.05 vs control; *p < 0.05 vs MASLD.mean and SEM.

Conclusion

Telmisartan improved high-fat diet–induced MASLD in rats, with benefits seen across metabolic measures, liver injury markers, oxidative stress, inflammation, mitochondrial function, and histology. Overall, the data support a mechanism where telmisartan helps by boosting mitochondrial performance (SIRT1/AMPK-related) and strengthening antioxidant defenses (Nrf2/HO-1), which together reduce liver damage and slow fibrotic progression.

Take-home message:
Telmisartan mitigates MASLD by improving mitochondrial function and antioxidant signaling, leading to less steatosis, inflammation, and fibrosis.

Implications / next steps:
- Supports telmisartan as a potential repurposing candidate for MASLD pathways beyond blood pressure control.
- Future work: dose–response, longer treatment, and validation in additional MASLD models (and eventual clinical translation).

Acknowledgements

Special thanks to the University of Kansas School of Pharmacy.

References

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